BUS EDS RECOGNITION PROJECT

# 01 Problem, Target User & Value

## Recognising bus EDS for a more inclusive Singapore

PUBLIC TRANSPORT FOR A MORE INCLUSIVE SINGAPORE

### The Problem

- Visually impaired commuters may struggle to identify an approaching bus quickly and independently.
- Electronic Display Signs (EDS) can be difficult to read due to distance, motion blur, glare, low light and viewing angle.
- Missing the bus number or destination can create uncertainty and reliance on others.

### Target User

- **Primary:** visually impaired commuters in Singapore.
- Also useful for low-vision and elderly commuters.
- **Potentially helpful** for passengers in unfamiliar transport environments (e.g. tourists, new residents).

### Proposed Value

- **Detect and recognise** bus EDS information automatically from video.
- **Provide a reliable** bus number, road and destination to the user.
- **Improve access** to bus service information, promoting greater independence in public transport.
- Support a more inclusive and accessible Singapore.




### Our Vision

A computer vision system that recognises Singapore bus EDS information and helps visually impaired commuters identify the right bus with greater independence.

# 02 Objectives & Scope

Clear goals and explicit boundaries

## CONFIRMED OBJECTIVES

1. Detect the EDS screen directly, without a separate bus-detection stage.
2. Crop accepted EDS regions and recognise service number, destination, and road names.
3. Use fuzzy matching against available bus reference data as the initial correction method.
4. Accept computer-camera streaming and uploaded videos.
5. Support low-latency processing; initially sample up to 15 frames per second.
6. Present results through a local web application.

## CURRENT SCOPE

- Singapore buses, no operator restriction
- Different EDS formats intended to be supported
- FastAPI backend
- HTML, CSS and JavaScript frontend
- Run on localhost

## OUTSIDE CURRENT SCOPE

- Audio / text-to-speech
- Mobile deployment
- Cloud deployment

## DEMONSTRATION TARGET

**INPUT**
Camera stream + uploaded video

→

**PROCESSING**
Initial sampling up to 15 FPS

→

**OUTPUT**
Low-latency text on localhost

# 03 High-Level System Workflow

**Offline preparation supports the runtime pipeline**

## OFFLINE PREPARATION

- Collect Singapore bus images via web-scraping / Google image search.
- Collect route reference information from Wikipedia.
- Annotate EDS boundaries.
- Prepare cropped EDS text images with verified transcriptions.
- Fine-tune detection and OCR candidates and compare baselines.
- Prepare the reference CSV and evaluate before selecting the final pipeline.

## RUNTIME SEQUENCE

**01 Camera / uploaded video**
Receive input

→

**02 Frame sampling**
Initially up to 15 FPS

→

**03 EDS detection**
Direct EDS detection plus user-defined threshold

→

**04 Crop EDS region**
Cropping required; extra preprocessing undecided

**05 OCR**
Extract requested text fields

→

**06 Fuzzy matching**
Compare with bus reference data

→

**07 Controlled output**
Display results at a chosen interval

FAILURE HANDLING

**Low OCR confidence → “No bus information detected”**

Keep detection confidence, OCR confidence, and fuzzy similarity separate; they measure different things.

# 04 Data, Models & Technology Stack

Separate confirmed decisions from candidates and open choices

## DATA

Sources and preparation

### Bus images

Singapore bus images discovered through Google image search using a scraping script.

### Reference data

Wikipedia route information plus bus information.csv.

### Detection labels

EDS boundary annotations required; format unconfirmed.

### OCR labels

Cropped EDS text images with verified text labels required.

### Cleaning

Check duplicates, unreadable images, missing values and inconsistent text.

## MODEL CANDIDATES

Potential options (not final)

### Detection

| | | |
|---|---|---|
| YOLO11n | First detector candidate | Recommended |
| YOLO11s | Speed vs accuracy comparison | Recommended |
| Faster R-CNN ResNet-50 FPN v2 | Optional family comparison | Optional |

### OCR

| | | |
|---|---|---|
| EasyOCR | Ready-to-use baseline | Recommended |
| TrOCR (printed-text) | Fine-tuned comparison candidate | Recommended |
| PaddleOCR PP-OCRv5 | Optional alternative | Optional |

## TECHNOLOGY DECISIONS

Tools and infrastructure

| | | |
|---|---|---|
| Backend | FastAPI | CONFIRMED |
| Frontend | HTML, CSS, JavaScript | CONFIRMED |
| Deployment | Localhost | CONFIRMED |
| Programming language | Python | IMPLIED |
| Deep learning | Pytorch | RECOMMENDED |
| Video/image handling | OpenCV | RECOMMENDED |
| Reference matching | Fuzzy matching | CONFIRMED INITIALLY |
| Annotation software | To be confirmed | OPEN |

# 05 Reference Matching & Local Application

Post-processing supports OCR; it does not guarantee ground truth

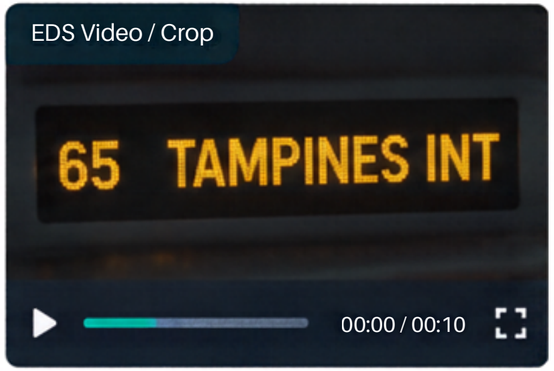
## REFERENCE MATCHING

1. **Raw OCR**
   Keep the model output for error analysis.
2. **Normalise**
   Clean text using agreed rules.
3. **Fuzzy similarity**
   Compare OCR text with busInformation.csv.
4. **Candidate selection**
   Select the closest candidate using the initial rule.
5. **Review effect**
   Measure fixes, harmful corrections and rejected valid results.

**CURRENT CSV LIMITATIONS**

- No separate road name field, no display-direction field, no EDS alias field.
- Origin and destination are route reference fields, not verified EDS transcriptions.

## LOCAL WEB APPLICATION




Raw OCR
**65 | Tamp1nes Int**

Reference-based result
**65 | Tampines Int**

OCR confidence: 0.xx

- Computer-camera streaming + uploaded videos
- FastAPI backend
- HTML, CSS and JavaScript frontend
- Continuing text updates at a controlled interval
- Run on localhost
- Audio is not part of the current implementation scope

**Detection confidence ≠ OCR confidence ≠ fuzzy similarity**

# 06 Evaluation Framework

## Measure detection, recognition, matching, and full-pipeline performance

EDS DETECTION

### mAP@0.5:0.95

Measures localisation and detection quality.

EDS DETECTION

### Recall at operating threshold

Measures missed EDS screens.

RAW OCR

### Character Error Rate (CER)

Lower is better; evaluate on verified text.

OCR / MATCH

### Field exact-match accuracy

Report raw OCR and post-match results separately.

END-TO-END

### Complete-result accuracy

All required visible fields must be correct.

LOCAL PERFORMANCE

### FPS + median / p95 latency

Time the complete capture-to-display pipeline.

## FAIR COMPARISONS

| | |
|---|---|
| Detector / training setting | Same held-out images, labels, resolution policy, hardware |
| Untouched vs fine-tuned OCR | Same verified crops and textnormalisation rules |
| Raw OCR vs + reference matching | Same predicted crops, frames, fields, and ground truth |
| Original vs transformed examples | Fixed transformations and severity settings |
| Complete live/upload pipeline | Same declared hardware and video settings |

## STATUS OF RESULTS

**No numerical results or target thresholds have been supplied yet.**

Show the evaluation plan only. Do not invent accuracy, FPS, progress percentages or targets.

# 07 Challenges & Next Steps

## Design risks, recommended responses, and future direction



### DESIGN RISKS & RESPONSES

Each risk is paired with a planned response

**1 — IMAGE QUALITY**
**Small / blurred / angled EDS text**

✓ **Response:** Collect representative examples; inspect crop quality; test OCR separately from detection.

**2 — REFERENCE DATA**
**Road-name reference coverage**

✓ **Response:** Decide whether to expand reference data or retain unmatched recognised road text.

**3 — MATCHING**
**Reference forcing / ambiguous endpoints**

✓ **Response:** Test harmful corrections; define rejection and direction rules.

**4 — PERFORMANCE**
**Runtime slower than sampled input**

✓ **Response:** Measure full latency; drop stale frames or reduce processing frequency if needed.

**5 — ACCESSIBILITY**
**Text-only interface**

✓ **Response:** Present as a CV prototype; audio and accessibility testing remain later work.

**6 — DATASET**
**Scraped data errors / duplicates**

✓ **Response:** Clean data, record sources and separate near-duplicates before evaluation.

### RECOMMENDED NEXT STEPS

**PREPARE**

1. Confirm annotation process and reference coverage.
2. Choose detector / OCR candidates.
3. Prepare verified detection and OCR labels.

**BUILD**

4. Establish baselines.
5. Fine-tune and compare.
6. Implement fuzzy matching and rejection behaviour.
7. Integrate camera / upload input and text output.

**EVALUATE & DEMO**

8. Evaluate accuracy, robustness, and latency.
9. Prepare the demonstration.

CONFIRMED FUTURE DIRECTION

**Expand to buses in other countries.**

New imagery · Labels · Display vocabularies · Route references · → Evaluation




**Not current commitments:** audio remains optional future work; mobile, cloud and other signage are not planned yet.